Resolutions and Hilbert's Syzygy Theorem

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Bibliography

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② *The geometry of syzygies*

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1 Free Resolutions

Definition 1.1. *A sequence* $M'' \xrightarrow{\psi} M \xrightarrow{\varphi} M'$ *is* <u>*exact*</u>*, if* $\operatorname{Im}\psi = \operatorname{Ker}\varphi$.

Definition 1.2. <u>*Short exact sequence*</u> *is a following exact sequence:*

$$0 \to M'' \to M \to M' \to 0$$

Remark 1.3. *For module homomorphism* $\varphi : M \to M'$*:*

i) φ *injective* $\iff$ $0 \to M \xrightarrow{\varphi} M'$ *exact*

ii) φ *surjective* $\iff$ $M \xrightarrow{\varphi} M' \to 0$ *exact*

Generally, we want to discuss the properties of a specific module M. Some algebraist figured out a way to treat M as a quotient, discuss.

Recall that every finitely generated module M has the form

$$M \cong F_0/K_0$$

where F_0 is a finitely generated free R-module and K_0 is an R-module. Since F_0 is finitely generated and R is Noetherian, the submodule $K_0 \subseteq F_0$ is also finitely generated.

This implies that we can write

$$K_0 \cong F_1/K_1$$

where F_1 is a finitely generated free R-module and K_1 is an R-module.

Since K_1 is again finitely generated, we can iterate this process. This constructs a sequence:

$$\cdots \to F_n \xrightarrow{\varphi_n} F_{n-1} \to \cdots \to F_1 \xrightarrow{\varphi_1} F_0 \to 0$$

where each map φ_n factors as $F_n \twoheadrightarrow F_n/K_n \cong K_{n-1} \hookrightarrow F_{n-1}$.

Definition 1.4. *A <u>free resolution</u> of M, not necessarily finitely generated, is the sequence of $\{F_k\}$, denoted by $\mathcal{F}$, s.t.*

$$\mathcal{F} : \cdots \to F_n \xrightarrow{\varphi_n} F_{n-1} \to \cdots \to F_1 \xrightarrow{\varphi_1} F_0$$

exact and $\operatorname{Coker} \varphi_1 \cong M$.

Definition 1.5. *Here $\varphi : M \to N$, the <u>cokernel</u> is defined by* $\operatorname{Coker} \varphi = N / \operatorname{Im} \varphi$.

Definition 1.6. *If $\exists n \in \mathbb{Z}^+$, s.t. $F_n = 0$ and $F_{n-1} \neq 0$, we say $\mathcal{F}$ is a <u>finite resolution of length n</u>.*

Definition 1.7. *We call* $\operatorname{Im} \varphi_k$ *<u>the k-th syzygies of M</u>.*

In the study of an unknown mathematical object, it's always a good approach to study some of its property, which becomes invariant under specific condition. In our specific case of module, we introduced "free resolution", from which various beautiful properties can be obtained.

Theorem 1.8 (Hilbert Syzygy Theorem)**.** *[Eisenbud, 1: Thm 1.13] For $R = k[x_1, \ldots, x_n]$, any finitely generated graded R-module has a finite free resolution*

$$0 \to F_m \xrightarrow{\varphi_m} F_{m-1} \to \cdots \to F_1 \xrightarrow{\varphi_1} F_0$$

Moreover, we may take $m \leq n$.

Proof. C.f. [Eisenbud, 1: Chap 15 & Chap 19.] □

Example 1.9 (Koszul Complex)**.**

$$\begin{aligned}
K(x_1): &\quad 0 \to R(-1) \xrightarrow{[x_1]} R \\
K(x_1, x_2): &\quad 0 \to R(-2) \xrightarrow{\left[\begin{smallmatrix} x_2 \\ -x_1 \end{smallmatrix}\right]} R^2(-1) \xrightarrow{[x_1 \ x_2]} R \\
K(x_1, x_2, x_3): &\quad 0 \to R(-3) \xrightarrow{\left[\begin{smallmatrix} x_1 \\ x_2 \\ x_3 \end{smallmatrix}\right]} R^3(-2) \xrightarrow{\left[\begin{smallmatrix} 0 & x_3 & -x_2 \\ -x_3 & 0 & x_1 \\ x_2 & -x_1 & 0 \end{smallmatrix}\right]} R^3(-1) \xrightarrow{[x_1 \ x_2 \ x_3]} R
\end{aligned}$$

Lemma 1.10 (Atiyah, 1: Proposition 2.11)**.** *Any commutative ring R, For exact sequence $0 \to M_n \to M_{n-1} \to \cdots \to M_1 \to 0$ of R-modules, for any additive function λ acting on R-modules, we have:*

$$\sum_{k=0}^{n} (-1)^k \lambda(M_k) = 0$$

Corollary 1.11 (Hilbert Functions)**.** *[Eisenbud, 2: Corollary 1.2] For $R = k[x_1, \ldots, x_n]$, graded R-module M. If M has finite free resolution*

$$0 \to F_m \to F_{m-1} \to \cdots \to F_1 \to F_0$$

where each F_k is finitely generated free module $F_k = \bigoplus_i S(-a_{k,i})$, then

$$H_M(d) = \sum_{k=0}^{m} (-1)^k \sum_i \binom{n + d - a_{k,i} - 1}{n - 1}$$

Proof. By Lemma 1.10, trivial. □

From last lecture, we know if d is "big enough", $H_M(d)$ will conform to a polynomial, namely "Hilbert polynomial". From Corollary 1.11, we may guarantee the following:

Corollary 1.12 (Eisenbud, 2: Corollary 1.3)**.** *From Corollary 1.11, if $d \geq \max\{1 + a_{i,j} - n\}$, $H_M(d) = P_M(d)$ conforms to the Hilbert Polynomial.*

2 Minimal Free Resolutions

Definition 2.1. *A complex of R-modules is a sequence $\{F_k\}_{k\in\mathbb{Z}}$ of R-modules*

$$\cdots \to F_{k+1} \to F_k \to F_{k-1} \to \cdots$$

such that $\forall k \in \mathbb{Z}$, compositions $F_{k+1} \to F_k \to F_{k-1}$ are all zero maps.

Definition 2.2. *The homology of this complex at F_k is*

$$H_k = \operatorname{Ker}(F_k \to F_{k-1}) / \operatorname{Im}(F_{k+1} \to F_k)$$

We say F_k doesn't have homology if $H_k = 0$.

Definition 2.3. *$R = k[x_1, \ldots, x_n]$, a complex of R-modules*

$$\cdots \to F_k \xrightarrow{\varphi_k} F_{k-1} \to \cdots$$

is minimal if for each k, $\operatorname{Im}\varphi_k \subseteq \mathfrak{m}F_{k-1}$. Here $\mathfrak{m} = (x_1, \ldots, x_n) = R\backslash k$ is the homogeneous maximal ideal, a.k.a, irrelevant ideal.

Lemma 2.4 (Graded Nakayama's Lemma)**.** *[Eisenbud, 2: Lemma 1.4] $R = k[x_1, \ldots, x_n]$, M is a finitely generated graded R-module, $\mathfrak{m}$ irrelevant ideal. Consider the canonical homomorphism $M \to M/\mathfrak{m}M$. For $m_1, \ldots, m_n \in M$, if $\langle \bar{m}_1, \ldots, \bar{m}_n \rangle = M/\mathfrak{m}M$, then $\langle m_1, \ldots, m_n \rangle = M$.*

Proof. Consider $\tilde{M} = M/\sum Rm_k$, we'll prove $\tilde{M} = 0$. Since $\{m_1, \ldots, m_n\}$ generate $\tilde{M}$, we have $\overline{\tilde{M}} = 0$, i.e.

$$\mathfrak{m}\tilde{M} = \tilde{M}$$

Suppose $\tilde{M} \neq 0$, then since it's finitely generated, we consider a least degree term in one of the set of its generators, denoted by $\tilde{m}$. Recall $\mathfrak{m} = R \setminus k$, so any term within has a positive degree. Then $\tilde{m} \notin \mathfrak{m}\tilde{M}$, contradictory! □

Remark 2.5. *See the general version of Nakayama's Lemma in [Atiyah, 1: Proposition 2.6].*

Corollary 2.6 (Eisenbud, 2: Corollary 1.5)**.** *A graded free resolution*

$$\mathcal{F} : \cdots \to F_n \xrightarrow{\varphi_n} F_{n-1} \to \cdots \to F_1 \xrightarrow{\varphi_1} F_0$$

is minimal as a complex iff $\forall k$, the map φ_k takes a basis of F_k to a minimal set of generators of $\operatorname{Im}\varphi_k$.

To prove this corollary, we need some Lemma first.

Lemma 2.7 (Atiyah, 1: Exercise 2.2)**.** *For ring R and its ideal I, let M be a R-module, then*

$$M/IM \cong (R/I) \otimes_R M$$

Proof. Trivial. See [Flandre71, 1: Chap 2]. □

Lemma 2.8 (Atiyah, 1: Proposition 2.18)**.** *Ring R and R-module N. If*

$$M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$$

is exact, then

$$M' \otimes N \xrightarrow{f\otimes 1} M \otimes N \xrightarrow{g\otimes 1} M'' \otimes N \to 0$$

is exact. Here 1 denotes the identity map.

Proof of Corollary 2.6. Consider right exact sequence $F_{k+1} \to F_k \to \operatorname{Im}\varphi_k \to 0$. By Lemma 2.8,

$$(R/\mathfrak{m}) \otimes_R F_{k+1} \to (R/\mathfrak{m}) \otimes_R F_k \to (R/\mathfrak{m}) \otimes_R \operatorname{Im}\varphi_k \to 0$$

is exact. By Lemma 2.7, such sequence is isomorphic to

$$F_{k+1}/\mathfrak{m}F_{k+1} \to F_k/\mathfrak{m}F_k \to \operatorname{Im}\varphi_k/\mathfrak{m}\operatorname{Im}\varphi_k \to 0$$

$\mathcal{F}$ is minimal iff $\forall k$, $F_{k+1}/\mathfrak{m}F_{k+1} \to F_k/\mathfrak{m}F_k$ is zero map, which is equivalent to say

$$F_k/\mathfrak{m}F_k \to \operatorname{Im}\varphi_k/\mathfrak{m}\operatorname{Im}\varphi_k$$

is an isomorphism. By Graded Nakayama's Lemma, this is equivalent to say a basis of F_k was mapped to a minimal generating set of $\operatorname{Im}\varphi_k$. □

As we said before, it's always desirable for us to look for invariants when exploring a mathematical concept.

Specifically, in the study of finitely generated graded module of $R = k[x_1, \ldots, x_n]$, we'll see minimal graded free resolution is unique under isomorphisms.

Theorem 2.9 (Uniqueness of minimal free resolution)**.** *[Eisenbud, 2: Theorem 1.6] For $R = k[x_1, \ldots, x_n]$, finitely generated graded R-module M,*

i) If $\mathcal{F}, \mathcal{G}$ are minimal free resolutions of M, then there exist a graded isomorphism of complex $\mathcal{F} \to \mathcal{G}$, inducing the identity map on M.

ii) Every free resolution of M, denoted by $\mathcal{H}$, we have

$$\mathcal{H} \cong \mathcal{F} \oplus \mathcal{H}_0$$

where $\mathcal{F}$ is a minimal free resolution, $\mathcal{H}_0$ is a trivial complex.

Proof. See [Eisenbud, 1: Theorem 20.2]. □

One thing worth noticing is that: For any minimal free resolution of a finitely generated graded R-module, the number of each degree required for each F_k only depends on M! See Theorem 2.12. Before that, we need a new definition.

Definition 2.10. *A* <u>Tor *functor*</u> *is a functor in categories of modules, acting by $N \otimes_R -$ where N is a random member. For R-module N and its free resolution*

$$\cdots \to F_n \xrightarrow{\varphi_n} F_{n-1} \to \cdots \to F_1 \to F_0 \to M \to 0$$

we have

$$\cdots \to F_n \otimes_R N \xrightarrow{\varphi_n \otimes 1} F_{n-1} \otimes_R N \to \cdots \to F_1 \otimes_R N \xrightarrow{\varphi \otimes 1} F_0 \otimes_R N \to 0$$

as a complex, and we define $\operatorname{Tor}_k^R(M, N) = \operatorname{Ker}(\varphi_k \otimes 1)/\operatorname{Im}(\varphi_{k+1} \otimes 1)$.

Remark 2.11. *The term "free resolution" above can be generalized into projective resolution.*

Theorem 2.12 (Eisenbud, 2: Proposition 1.7)**.** *(Its proof has typo!) $R = k[x_1, \ldots, x_n]$, finitely generated graded R-module M and its minimal free resolution $\mathcal{F}$, irrelevant ideal $\mathfrak{m}$. $\mathbb{K} = R/\mathfrak{m}$ is the residue field, then any minimal set of homogeneous generators of F_k contains exactly*

$$\operatorname{Dim}_{\mathbb{K}} \operatorname{Tor}_k^R(\mathbb{K}, M)_i$$

generators of degree i.

Proof. $\operatorname{Tor}_k^R(\mathbb{K}, M)_i$ is the degree i component of the graded vector space, which is the k-th homology of $\mathbb{K} \otimes_R \mathcal{F}$.

Since $\mathcal{F}$ is minimal, by the definition of minimal we know every maps in $\mathbb{K} \otimes_R \mathcal{F}$ is zero map. (Recall that minimality resulting in mapping into $\mathfrak{m}F_k$). Therefore

$$\begin{aligned}\operatorname{Tor}_k^R(\mathbb{K}, M) &= \operatorname{Ker}(\varphi_k \otimes 1)/\operatorname{Im}(\varphi_{k+1} \otimes 1) \\ &= \mathbb{K} \otimes_R F_k/0 \\ &= \mathbb{K} \otimes_R F_k \\ &\cong F_k/\mathfrak{m}F_k\end{aligned}$$

and by graded Nakayama's Lemma, $\operatorname{Dim}_{\mathbb{K}} \operatorname{Tor}_k^R(\mathbb{K}, M)_i$ is the number of i-degree generators that F_k required. □

3 Projective Dimension and Global Dimension

Definition 3.1. *An* <u>*epimorphism*</u> $\varphi : X \twoheadrightarrow Y$ *in a category is a morphism such that for any object Z, there exist at most 1 morphism in* $\operatorname{Mor}(Y, Z)$*, s.t.*

$$\begin{tikzcd} X \arrow[r, two heads, "\varphi"] \arrow[dr] & Y \arrow[d, dashed, "\leq 1"] \\ & Z \end{tikzcd}$$

commute. i.e., φ satisfy the right cancellation law.

Definition 3.2. *A* <u>*monomorphism*</u> $\varphi : X \hookrightarrow Y$ *is the case in Def 3.1, but all arrows are reversed. i.e. φ satisfy the left cancellation law.*

Remark 3.3. *All surjections are epimorphisms, all injections are monomorphisms.*

Definition 3.4. *A module P is projective if for every epimorphism of module $\alpha : M \twoheadrightarrow N$, and every map $\beta : P \to N$, $\exists \gamma : P \to M$, s.t. $\beta = \alpha\gamma$, i.e.,*

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists\gamma"'] \arrow[d, "\beta"] \\ M \arrow[r, two heads, "\alpha"'] & N \end{tikzcd}$$

commute.

Definition 3.5. *Ring R, a* <u>*projective resolution*</u> *of R-module is a complex*

$$\mathcal{P} : \cdots \to P_n \xrightarrow{\varphi_n} P_{n-1} \to \cdots \to P_1 \xrightarrow{\varphi_1} P_0$$

of projective R-module, s.t. $M = \operatorname{Coker} \varphi_1$ and $\mathcal{P}$ is exact. Its <u>*length*</u> *is defined in the same way as the one in free resolutions.*

Definition 3.6. *Ring R, we define the* <u>*projective dimension*</u> *of a R-module M, denoted by $\operatorname{Pd}_R M$, to be the minimum of lengths of projective resolutions ($\operatorname{Pd}_R M = \infty$ if M has no finite projective resolution).*

Definition 3.7. *A ring is* <u>*local*</u> *if it only has one maximal ideal.*

Remark 3.8. *$R = k[x_1, \ldots, x_n]$ is graded local, since it has the unique maximal ideal, namely the irrelevant ideal.*

Theorem 3.9. *Local ring R, $\mathbb{K} = R/\mathfrak{m}$ is the residue field, M is finitely generated nonzero R-module, then: $\forall k > \operatorname{Pd}_R M$, $\operatorname{Tor}_k^R(M, \mathbb{K}) = 0$.*

Proof. Trivial.

$$k > \operatorname{Pd}_R M \implies F_k = 0 \implies \operatorname{Tor}_k^R(M, \mathbb{K}) = \operatorname{Ker} \varphi_k / \operatorname{Im} \varphi_{k+1} = 0$$

□

Theorem 3.10 (Eisenbud, 2: Corollary 1.8)**.** *$R = k[x_1, \ldots, x_n]$, M is finitely generated graded R-module. Then*

$$\textit{length of the minimal free resolution} = \textit{projective dimension}$$

Proof. i) On one hand, a free resolution is projective resolution, so

$$\text{length of minimal free resolution} \geq \text{projective dimension}$$

ii) On the other hand, $\forall k > \operatorname{Pd}_R M$, $\operatorname{Tor}_k^R(\mathbb{K}, M) = 0$, meaning $F_k = 0$. Hence

$$\text{length of minimal free resolution} \leq \text{projective dimension}$$

Done. □

Definition 3.11. *The <u>global dimension</u> of ring R, denoted by $\operatorname{GLdim} R$, is the supremum of the projective dimensions of all R-modules.*

Lemma 3.12 (Eisenbud, 1: Theorem 19.1)**.** *On ring R, the following are equivalent:*

i) $\operatorname{GLdim} R \leq n$

ii) *$\forall R$-module M,* $\operatorname{Pd} M \leq n$

iii) *$\forall$ finitely generated R-module M,* $\operatorname{Pd} M \leq n$

Proof. See [Eisenbud, 1: Theorem A3.18]. □

Corollary 3.13. *$\forall$ local ring R, $\mathbb{K} = R/\mathfrak{m}$ being its residue field, we have*

$$\operatorname{GLdim} R = \operatorname{Pd}_R \mathbb{K}$$

Proof. $\forall$ finitely generated graded R-module M, we have $\forall k \in \mathbb{Z}^+$,

$$\operatorname{Tor}_k^R(\mathbb{K}, M) \cong \operatorname{Tor}_k^R(M, \mathbb{K})$$

So we may compute $\operatorname{Tor}_k^R(\mathbb{K}, M)$ from a projective resolution of $\mathbb{K}$ as well.

Therefore by Theorem 3.9, $\forall k > \operatorname{Pd}_R \mathbb{K}$, $\operatorname{Tor}_k^R(M, \mathbb{K}) = 0$. Hence $\operatorname{Pd}_R \mathbb{K} \geq \operatorname{Pd}_R M$. Since M was arbitrarily chosen, by Lemma 3.12, $\operatorname{Pd}_R \mathbb{K} \geq \operatorname{GLdim} R$

$$\implies \operatorname{Pd}_R \mathbb{K} = \operatorname{GLdim} R.$$

□

Remark 3.14. *(3.9), (3.10), (3.13) was put together in [Eisenbud, 1: Corollary 19.5] and called "homological version of Nakayama's Lemma".*

The general Nakayama's Theorem states that: for local ring R and its unique maximal ideal $\mathfrak{m}$, and residue field $\mathbb{K}$, we have

$$M/\mathfrak{m}M = 0 \implies M = 0$$

Notice $M/\mathfrak{m}M \cong \mathbb{K} \otimes_R M \cong \mathrm{Tor}_0^R(\mathbb{K}, M)$. So we can restated the General Nakayama's Lemma as:

$$\textit{The module } M \textit{ vanish iff } \mathrm{Tor}_0^R(\mathbb{K}, M) = 0.$$

On the other hand, we know a free resolution is minimal iff every maps (differentials) maps a basis of the free module to a minimal generators of its image, a.k.a. the k-th syzygies.

Therefore the homological version (which turns out to be a more general version of the general nakayama's Lemma) tells us that

$$\textit{The } (k+1)\textit{-th syzygy of } M \textit{ vanish iff } \mathrm{Tor}_{k+1}^R(\mathbb{K}, M) = 0.$$